CS341 · COMPUTER GRAPHICS

Midterm 462 Practice

20 marks · 1 hour

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01 Questions

Q1 FILL IN THE GAPS

1. Boundary representations (B-Reps) define the surface of 3D objects using piecewise smooth surfaces like _____ or _____.

2. Scan conversion is the process of converting _____ representations of shapes (e.g., lines, circles) into _____ on a raster display.

3. In Bresenham's line algorithm, the decision variable determines whether the direction of the next pixel is _____ or _____.

Q2-A TRUE / FALSE

1. Bresenham's line algorithm uses floating-point arithmetic to draw lines. ( )
2. In raster displays, pixels are arranged in a rectangular grid. ( )
3. Super-sampling is a method for anti-aliasing in computer graphics. ( )
4. A voxel model is an example of a boundary representation. ( )
5. A triangle mesh is preferred in B-Reps. ( )
6. Computer graphics applications are used only in the entertainment industry. ( )

Q2-B MCQ

1. Which of the following is an example of a geometric transformation that alters the size of the object?
   a) Translation b) Rotation c) Scaling d) Blurring
2. _____ is a model heavily used in medical imaging and fluid dynamics.
   a) B-Reps b) CSG c) Voxel d) Triangle Mesh
3. The decision variable in Bresenham's line algorithm determines:
   a) The slope of the line b) The next pixel to plot c) The color of the pixel d) The resolution of the line
4. The implicit equation of a circle is defined as:
   a) $F(x, y) = x^2 + y^2 - r^2$ b) $F(x, y) = mx + n$ c) $F(x, y) = ax + by + c$ d) $F(x, y) = x + y - r$

Q2-B MCQ · CONT.

5. In transformation matrices, a negative scaling factor is used for:
   a) Translation b) Rotation c) Mirroring d) Shearing
6. What is Constructive Solid Geometry (CSG) used for?
   a) Creating smooth surfaces
   b) Combining simple shapes using Boolean operations
   c) Representing voxel models
   d) Partitioning space into octrees
7. What is Binary Space Partitioning (BSP) used for?
   a) Dividing space into half-spaces using planes
   b) Creating voxel models
   c) Rendering implicit surfaces
   d) Defining polygonal meshes

Q3 SHORT ANSWER · 5 PTS

A. State two advantages of using triangle meshes in 3D object representation. [2 marks]

B. What are the two main types of 3D object representations? Explain briefly. [2 marks]

C. Name two applications of scan conversion in computer graphics. [1 mark]

Q4-A TRANSFORMATIONS · 2.5 PTS

Using concatenated transformations: rotate the point P = (5, 3) by 90 degrees around the point (1, 1).

Q4-B BRESENHAM · 3 PTS

Based on your knowledge about Bresenham's algorithm: generate the coordinates of the pixels that lie on a line segment having the endpoints (3, 3) and (10, 7) and draw it. (Show the steps of the solution.)

k	p_k	Decision	Next pixel	p_{k+1}
0				
1				
2				
3				
4				
5				
6				

